STA4001

Midterm Review

Acknowledgement: This PowerPoint is made by the last year's speaker Tang Yang, and modified by Me (Jian Muqing)

03/20/2025



主要内容

- 条件期望
- DTMC（离散时间马尔科夫链）

条件期望

$$E[g(X)|Y=y]=\frac{E[g(X)1(Y=y)]}{P(Y=y)}=\frac{\sum_{x\in\mathcal{S}_X}g(x)P(X=x,Y=y)}{\sum_{x\in\mathcal{S}_X}P(X=x,Y=y)}.$$

$$E[g(X)|Y=y]=\frac{E[g(X)1(Y=y)]}{P(Y=y)}=\frac{\int_{-\infty}^{\infty}g(x)f(x,y)dx}{\int_{-\infty}^{\infty}f(x,y)dx}.$$

例题

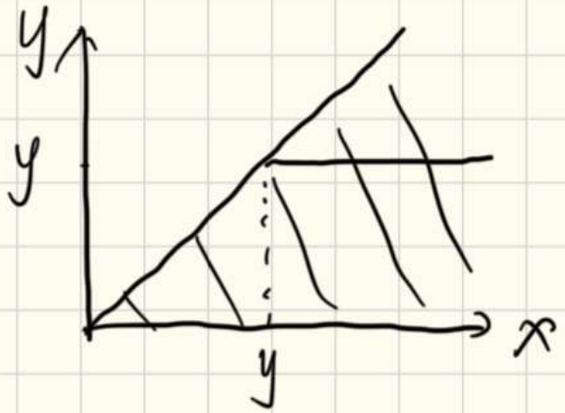
Example 3.7 Suppose the joint density of X and Y is given by

$$f(x,y) = \begin{cases} 4y(x-y)e^{-(x+y)}, & 0 < x < \infty, 0 \leqslant y \leqslant x \\ 0, & \text{otherwise} \end{cases}$$

Compute $E[X|Y = y]$.

例题

第1步：画出坐标系



第2步：求条件期望

$$E[X|Y=y] = \frac{E[X\mathbb{1}(Y=y)]}{Pr(Y=y)}$$

$$= \frac{\int_y^\infty x \cdot f(x,y)\,dx}{\int_y^\infty f(x,y)\,dx} = \cdots = \frac{(y+2)e^{-2y}}{e^{-2y}} = y+2$$

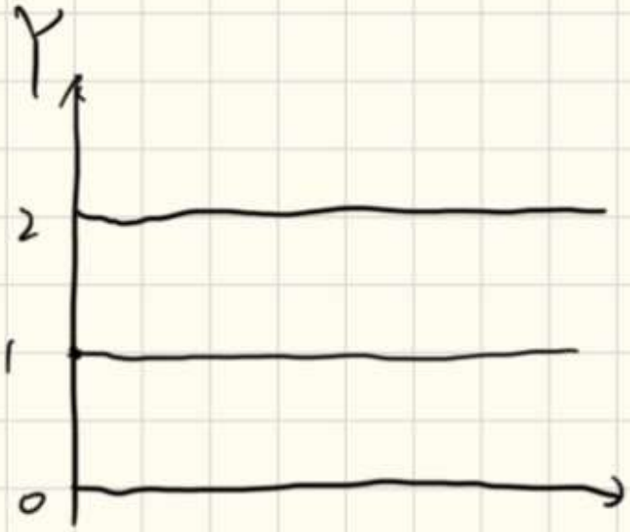
例题

Example 3.8 The joint density of X and Y is given by

$$f(x,y) = \begin{cases} \frac{1}{2}ye^{-xy}, & 0 < x < \infty, 0 < y < 2 \\ 0, & \text{otherwise} \end{cases}$$

What is $E[e^{X/2}|Y = 1]$?

例题



$$E[e^{\wedge}(\tfrac{X}{2}) \mid Y=1] = \frac{E[e^{\wedge}(\tfrac{X}{2})\,\mathbb{1}(Y=1)]}{P_r(Y=1)} = \frac{\int_0^{\infty} e^{\frac{x}{2}} f(x,1)\,dx}{\int_0^{\infty} f(x,1)\,dx}$$

例题

Let X_1 and X_2 be independent Poisson random variables with rates $\mu 1$ and $\mu 2$. Compute $E[X1 \mid X1 + X2 = t]$.

例题

$X_1 \sim Poi(\mu_1)$ $X_2 \sim Poi(\mu_2)$ $X_1 + X_2 \sim Poi(\mu_1+\mu_2)$

$$E[X_1 | X_1+X_2=t] = \frac{E[X_1 \mathbb{1}(X_1+X_2=t)]}{Pr(X_1+X_2=t)} = \frac{\sum_{x=0}^{t} x\, Pr(X_1=x) Pr(X_2=t-x)}{\cdots}$$

方法论

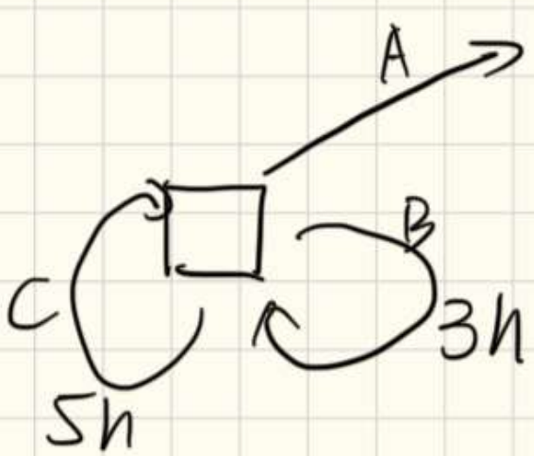
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- $$E[X] = E[E[X|Y]]$$
- $$\mathsf{Var}(X) = E[\mathsf{Var}(X|Y)] + \mathsf{Var}(E[X|Y]).$$

例题

Example 3.12: A miner is trapped in a mine containing three doors. The first door leads to a tunnel that takes him to safety after two hours of travel. The second door leads to a tunnel that returns him to the mine after three hours of travel. The third door leads to a tunnel that returns him to his mine after five hours. Assuming that the miner is at all times equally likely to choose any one of the doors, what is the expected length of time until the miner reaches safety?

例题

第1步分析：根据第1次的情况来分类讨论



$$E(T)=E[T\mid 选A]\cdot Pr(选A)+E[T\mid 选B]Pr(选B)$$
$$+E(T\mid 选C)\cdot Pr(选C)$$
$$=\frac{1}{3}\cdot 2+\frac{1}{3}(3+E(T))+\frac{1}{3}(5+E(T))$$

例题

A coin that comes up heads with probability p is continually flipped until the pattern T, T, H appears. (That is, you stop flipping when the most recent flip lands heads, and the two immediately preceding it lands tails.) Let X denote the number of flips made, and find $E[X]$.

例题

$X = Y + N$ 剩余步数.

Y：首次出现TT

N：

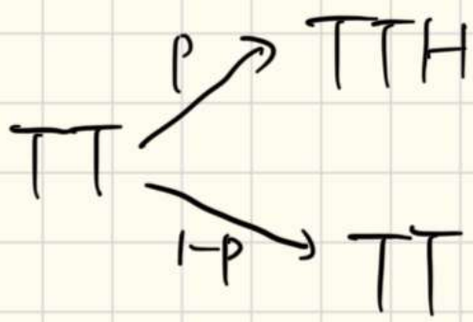
T T N

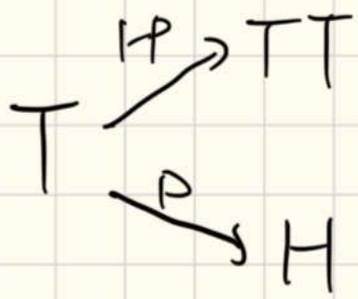
仍然用 first step analysis

$E(N) = p \cdot 1 + (1-p)\cdot(1+E(N))$

$E(N) = \frac{1}{p}$

$Y = Y_1 + Y_2$





$Y_2 = (1-p)\cdot 1 + p\cdot(1+Y_1+Y_2)$

$Y_1 = \frac{1}{1-p}$

例题

24. A coin, having probability p of landing heads, is continually flipped until at least one head and one tail have been flipped.
    (a) Find the expected number of flips needed.
    (b) Find the expected number of flips that land on heads.
    (c) Find the expected number of flips that land on tails.
    (d) Repeat part (a) in the case where flipping is continued until a total of at least two heads and one tail have been flipped.

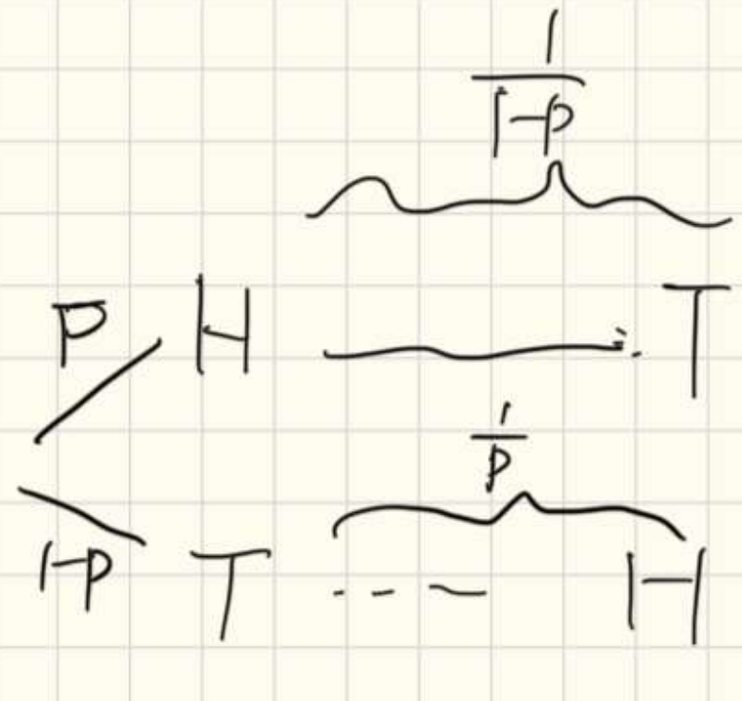
例题

仍然用 first step analysis

(a) $p\cdot\left(1+\frac{1}{1-p}\right)+(1-p)\cdot\left(1+\frac{1}{p}\right)$

(b) $p\cdot\left(1+\frac{1}{1-p}-1\right)+(1-p)\cdot 1$

(c) $p(1)+(1-p)\cdot\left(\frac{1}{p}\right)$



(a) $X = p\cdot X_1 + (1-p)X_2$

$= p\cdot X_1 + (1-p)\cdot\frac{2}{p}$

X_1：出现 1H1T

X_2：出现 2H

马尔可夫链

- Basics
- Long-run behavior
- Transient behavior
- Time Reverse
- Application

基本介绍

Defintion

- Consider a discrete-time stochastic process $\{X_n : n = 0, 1, 2, ..., \}$, taking values from $\{0, 1, 2, ...\}$.
- It is nature that the distribution of X_{n+1} should depend on the history X_n, X_{n-1},...,X_0.
- If the distribution of X_{n+1} depends *only* on X_n for all $n \geq 0$, we say the process X_n is a Markov chain. In detail,

$$P(X_{n+1} = j | X_n = i, X_{n-1} = i_{n-1}, ..., X_0 = i_0) = P(X_{n+1} = j | X_n = i) = P_{ij}.$$

- The value P_{ij} is called the one-step transition probability from state i to state j. Obviously,

$$P_{ij} \geq 0, \sum_{j=0}^{\infty} P_{ij} = 1, \text{ for all } i, j \geq 0.$$

The matrix $\mathbf{P} = (P_{ij})$ is called the transition probability matrix of X_n.

These equations are called Chapman-Kolmogorov equations. By these equations, we can prove that

$$P_{ij}^n = \mathbf{P}^{(n)}(i,j), \text{for all } n \geq 1.$$

构造state

- **Example 4.4** Suppose that whether or not it rains today depends on previous weather conditions through the last two days. Specifically, suppose that if it has rained for the past two days, then it will rain tomorrow with probability 0.7; if it rained today but not yesterday, then it will rain tomorrow with probability 0.5; if it rained yesterday but not today, then it will rain tomorrow with probability 0.4; if it has not rained in the past two days, then it will rain tomorrow with probability 0.2.

- If we let the state at time n depend only on whether or not it is raining at time n, then the preceding model is not a Markov chain (why not?).
- However, we can transform this model into a Markov chain by saying that the state at any time is determined by the weather conditions during both that day and the previous day.

构造state

#3 **Pattern's Appearing Time** Let $\{X_n, n \geq 0\}$ be a Markov chain with states 0,1,2,3. Suppose this Markov chain is irreducible and the transition probability is $P_{i,j} > 0,\ i, j = 0, 1, 2, 3$. Let N be the number of transitions, starting from the state 0, until the pattern 1,2,1,2 appears. That is,

$$N = \min\{n \geq 4 : X_{n-3} = 1, X_{n-2} = 2, X_{n-1} = 1, X_n = 2\}.$$

We want to evaluate $\mathbb{P}(N \leq k)$. (The key point is also to find a good Markov chain with only one absorbing state.)

构造state

Solution: Pattern's Appearing Time Similar to question #2, we need to construct a new Markov chain with an absorbing state. Notice that there are four steps to reach the pattern $1, 2, 1, 2$. Once we are in state 0 or state 3, the procedure has to restart. If $X_n = 2$ but $X_{n-1} \neq 1$, the procedure also needs to restart. So in total, we need 7 states as follows.

1. $Y_n = 1$, if $X_n = 1$ and $(X_{n-2}, X_{n-1}) \neq (1, 2)$.
2. $Y_n = 2$, if $X_n = 2, X_{n-1} = 1$ and the pattern $1, 2, 1, 2$ has not happened yet.
3. $Y_n = 3$, if $X_{n-2} = 1$, $X_{n-1} = 2$ and $X_n = 1$.
4. $Y_n = 4$, if pattern $1, 2, 1, 2$ has happened by time n.
5. $Y_n = 5$, if $X_n = 2$ and $X_{n-1} \neq 1$.
6. $Y_n = 6$, if $X_n = 0$.
7. $Y_n = 7$, if $X_n = 3$.

Clearly, state 4 is the absorbing state. The transition probability matrix of $\{Y_n\}$ is

$$\boldsymbol{Q} = \begin{pmatrix} P_{1,1} & P_{1,2} & 0 & 0 & 0 & P_{1,0} & P_{1,3} \\ 0 & 0 & P_{2,1} & 0 & P_{2,2} & P_{2,0} & P_{2,3} \\ P_{1,1} & 0 & 0 & P_{1,2} & 0 & P_{1,0} & P_{1,3} \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ P_{2,1} & 0 & 0 & 0 & P_{2,2} & P_{2,0} & P_{2,3} \\ P_{0,1} & 0 & 0 & 0 & P_{0,2} & P_{0,0} & P_{0,3} \\ P_{3,1} & 0 & 0 & 0 & P_{3,2} & P_{3,0} & P_{3,3} \end{pmatrix}.$$

Because $X_0 = 0$, i.e. $Y_0 = 6$, $\mathbb{P}(N \leq k) = \boldsymbol{Q}^k{}_{6,4}$.

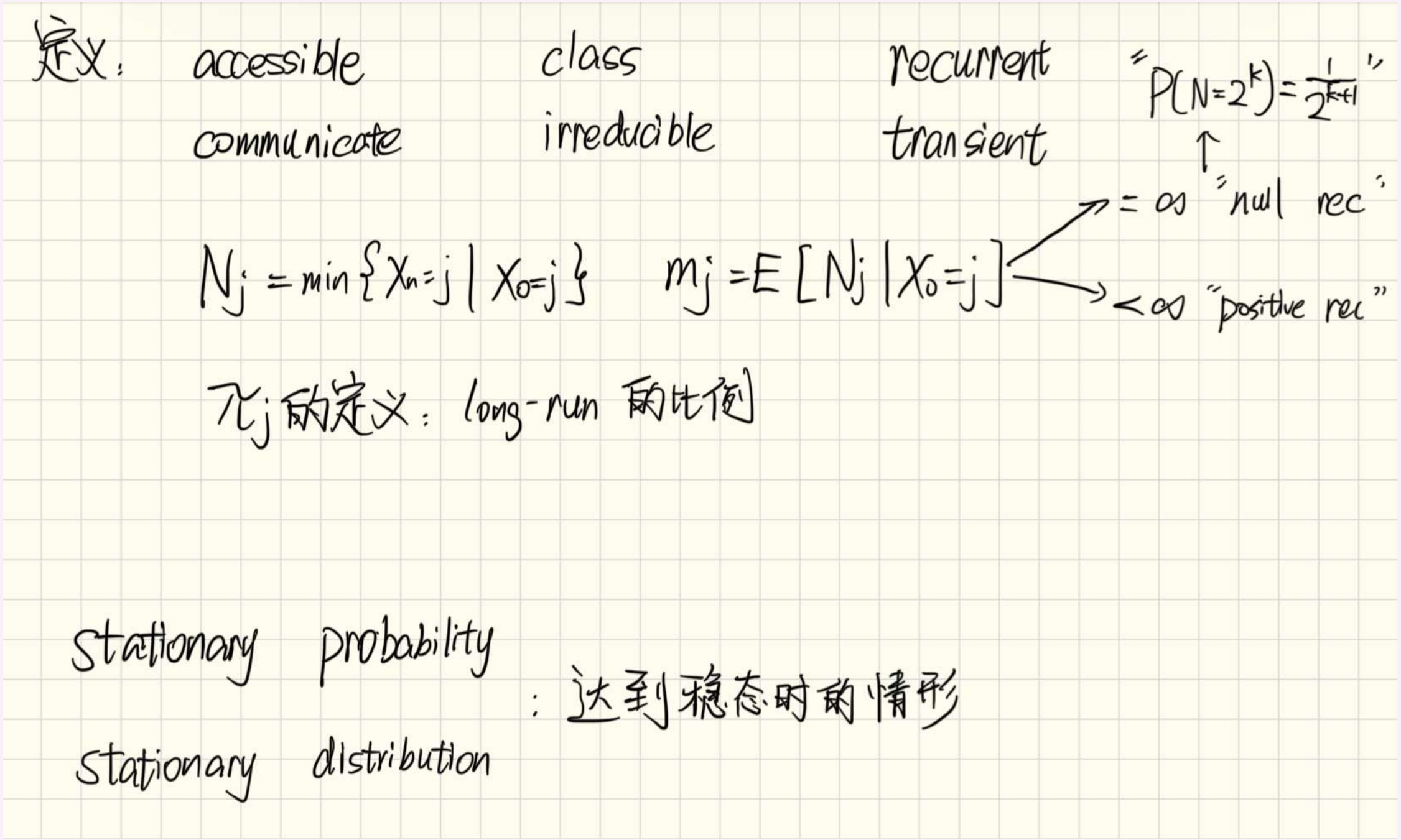
Absorbing Probability

- Consider a Markov chain with transition probabilities P_{ij}. Let $\mathcal{A}$ be a set of states, and suppose we are interested in the probability that the Markov chain ever enters any of the states in $\mathcal{A}$ by time m.That is,for a given state $i \notin \mathcal{A}$,we are interested in determining

$$\beta = P(X_k \in \mathcal{A} \text{ for some } k = 1, ..., m | X_0 = i)$$

- To determine the preceding probability we will define a Markov chain $\{W_n : n \geq 0\}$ whose states are the states that are not in $\mathcal{A}$ plus an additional state, which we will call $\mathcal{A}$. Once the W_n enters state $\mathcal{A}$ it remains there forever.

Long-run behavior



罕见特性：周期性 → 对 state i 来说

周期为 d 时，只要 n 不是 d 的倍数，$P_{ii}^{n}=0$

aperiodic：非周期性（周期为1）

当 $n=i, i+1, i+2, \ldots$ 时，P_{ii}^{n} 都 $\neq 0$，则一定说明非周期性

$$P=\begin{array}{c} \\ 1 \\ 2 \\ 3 \end{array}\begin{array}{c} \begin{array}{ccc} 1 & 2 & 3 \end{array} \\ \begin{bmatrix} 0 & 0.7 & 0.3 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix} \end{array}$$

$P_{11}^{2n+1}=0 \Rightarrow$ 说明 state 1 周期为2

$P_{22}^{2n+1}=0 \Rightarrow$ 说明 state 2 周期为2

1. Let $Z_0, Z_1, Z_2, \ldots$ be independent, identically distributed random variables with common p.m.f.:

k	0	1	2	3
p_k	0.1	0.3	0.2	0.4

For $n \geq 0$, define $X_n = \min(Z_0, Z_1, \ldots, Z_n)$.

(a) Give the transition matrix P of the Markov chain (X_n). **[5 marks]**

(b) Calculate $\mathbb{P}(X_3 = 2, X_2 = 1, X_1 = 2 | X_0 = 3)$ **[3 marks]**

(c) Calculate $\mathbb{P}(X_3 = 0, X_2 = 1, X_1 = 2 | X_0 = 3)$. **[3 marks]**

(d) Find the communication classes of the chain. **[3 marks]**

(e) Find the recurrent and transient states of the chain. **[4 marks]**

(f) What are the periods of the states? **[4 marks]**

(g) Does the chain have a stationary (invariant) distribution? **[3 marks]**

(h) Find the limiting probabilities of the chain (if any). **[5 marks]**

Solution.

(a) We have $X_n = \min(Z_0, Z_1, \ldots, Z_n), n \geq 1$
The transition matrix is shown as follows:

$$\mathbf{P} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0.1 & 0.9 & 0 & 0 \\ 0.1 & 0.3 & 0.6 & 0 \\ 0.1 & 0.3 & 0.2 & 0.4 \end{bmatrix}$$

(b) $\mathbb{P}(X_3 = 2, X_2 = 1, X_1 = 2 | X_0 = 3) = P_{32}P_{21}P_{12} = 0$

(c) $\mathbb{P}(X_3 = 0, X_2 = 1, X_1 = 2 | X_0 = 3) = P_{32}P_{21}P_{10} = 0.2 * 0.3 * 0.1 = 0.006$

(d) Since $X_m \leq X_{m-1} (\forall m \geq 1)$,
$i \to j$ is impossible if $i < j$.
Therefore, all classes singleton and we have 4 classes: $0, 1, 2, 3$

(e) For $i = 1, 2, 3, P_{i,i-1} > 0$ and once at $i - 1$, the MC cannot return to i.
Therefore $f_i < 1$ and i is transient.
For $i = 0$ is absorbing, thus recurrent.

(f) For all i, $p_{ii} > 0$, so "i" is aperiodic.

(g) For $\pi = (\pi_0, \pi_1, \pi_2, \pi_3)$ Solving $\pi P = \pi$, i.e.

$$\begin{cases} \pi_0 = \pi_0 + 0.1\pi_1 + 0.1\pi_2 + 0.1\pi_3 \\ \pi_1 = 0.9\pi_1 + 0.3\pi_2 + 0.3\pi_3 \\ \pi_2 = 0.6\pi_2 + 0.2\pi_3 \\ \pi_3 = 0.4\pi_3 \end{cases}$$

From the equations: $\pi_3 = 0$, then $\pi_2 = 0, \pi_1 = 0, \pi_0 = 1$
There is one unique stationary distribution$\pi = (1, 0, 0, 0)$

(h) Since $j = 1, 2, 3$ are transient,
$P_{ij}^n \to 0$ as $n \to \infty$ for all $0 \leq i \leq 3$ and $1 \leq j \leq 3$
Moreover,

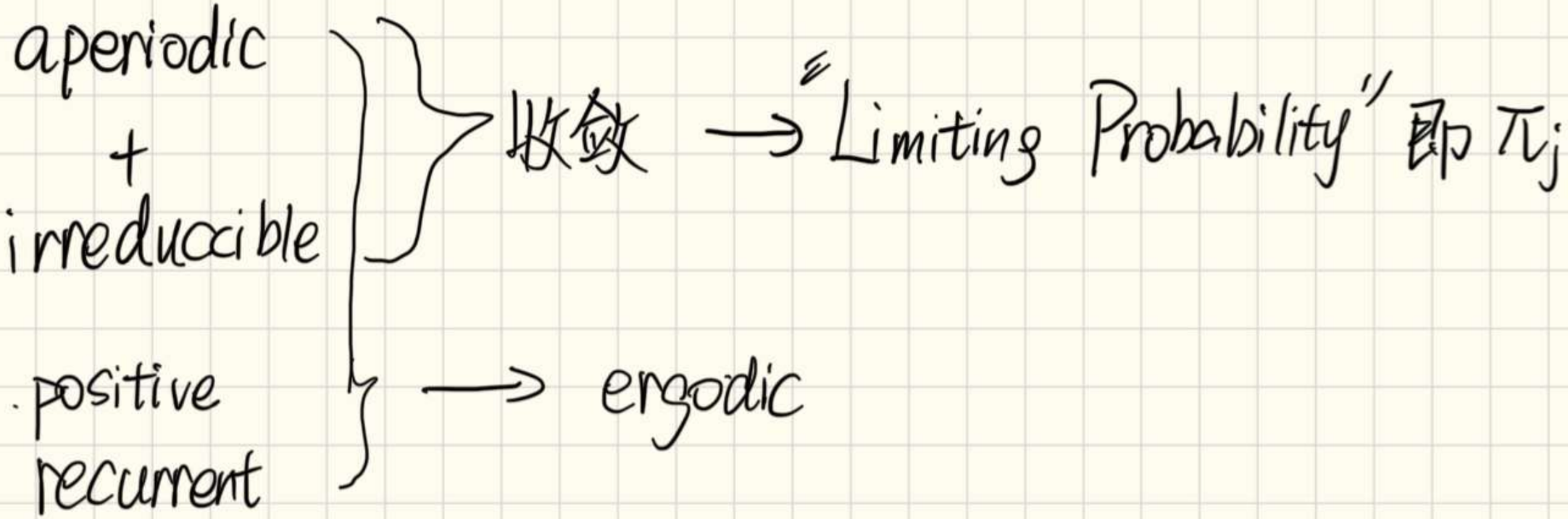
$$1 = P_{i0}^n + P_{i1}^n + P_{i2}^n + P_{i3}^n$$

implies $P_{i0}^n \to 1 (\forall 0 \leq i \leq 3)$
Therefore, as $n \to \infty$

$$P^n \to \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}$$

which are the limiting probability of the MC.

为
因


aperiodic
+
irreducible
positive
recurrent
收敛 → "Limiting Probability" 即 π_j
→ ergodic

例题

抽卡游戏：
每抽一次可以获得3星或4星角色，基础出货率10%，第5抽开始每抽增加10%概率，十发必出4星。求4星角色的综合出货率。

例题

抽卡游戏：
每抽一次可以获得3星或4星角色，基础出货率10%，
第5抽开始每抽增加10%概率，十发必出4星.
求4星角色的综合出货率。

考虑马尔可夫链

state i
→ 垫了i抽

$\pi_0 = 17.28\%$

	0	1	2	3	4	5	6	7	8	9
0	0.1	0.9	0	0	0	0	0	0	0	0
1	0.1		0.9							
2	0.1			0.9						
3	0.1				0.9					
4	0.1					0.9				
5	0.2						0.8			
6	0.3							0.7		
7	0.4								0.6	
8	0.5									0.5
9	1									

Mean Time Spent in Transient States

- Consider a finite-state Markov chain with transient states $T = \{1, 2, ..., t\}$. Let

$$\mathbf{P}_T = \begin{pmatrix} P_{11} & P_{12} & ... & P_{1t} \\ . & . & . & . \\ . & . & . & . \\ . & . & . & . \\ P_{t1} & P_{t2} & ... & P_{tt} \end{pmatrix}$$

- Suppose s_{ij} is the expected number of times that the Markov chain is in state j given that it starts in state i:

$$s_{ij} = E\left[\sum_{n=0}^{\infty} 1(X_n = j)|X_0 = i\right].$$

Let $\mathbf{S} = (s_{ij})_{1 \leq i,j \leq t}$, then

$$\mathbf{S} = I + \mathbf{P}_T\mathbf{S} \;\Rightarrow\; \mathbf{S} = (I - \mathbf{P}_T)^{-1}.$$

例题

抽卡游戏：
每抽一次可以获得3星或4星角色，基础出货率10%，
第5抽开始每抽增加10%概率，十发必出4星。
现在你已经垫了6抽，请问你期望还有多少抽出4星？

对马尔可夫链微调，

让 state 0 成为吸收态

$$P = \begin{array}{c} \\ 0 \\ 1 \\ 2 \\ 3 \\ 4 \\ 5 \\ 6 \\ 7 \\ 8 \\ 9 \end{array} \begin{array}{c} \begin{array}{cccccccccc} 0 & 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 \end{array} \\ \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0.1 & 0 & 0.9 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0.1 & 0 & 0 & 0.9 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0.1 & 0 & 0 & 0 & 0.9 & 0 & 0 & 0 & 0 & 0 \\ 0.1 & 0 & 0 & 0 & 0 & 0.9 & 0 & 0 & 0 & 0 \\ 0.2 & 0 & 0 & 0 & 0 & 0 & 0.8 & 0 & 0 & 0 \\ 0.3 & 0 & 0 & 0 & 0 & 0 & 0 & 0.7 & 0 & 0 \\ 0.4 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0.6 & 0 \\ 0.5 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0.5 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \end{array}$$

$$P_T = \begin{array}{c} \\ 6 \\ 7 \\ 8 \\ 9 \end{array} \begin{array}{c} \begin{array}{cccc} 6 & 7 & 8 & 9 \end{array} \\ \begin{bmatrix} 0 & 0.7 & 0 & 0 \\ 0 & 0 & 0.6 & 0 \\ 0 & 0 & 0 & 0.5 \\ 0 & 0 & 0 & 0 \end{bmatrix} \end{array}$$

$$S = (I - P_T)^{-1} = \begin{array}{c} \\ 6 \\ 7 \\ 8 \\ 9 \end{array} \begin{array}{c} \begin{array}{cccc} 6 & 7 & 8 & 9 \end{array} \\ \begin{bmatrix} 1 & 0.7 & 0.42 & 0.21 \\ 0 & 1 & 0.6 & 0.3 \\ 0 & 0 & 1 & 0.5 \\ 0 & 0 & 0 & 1 \end{bmatrix} \end{array}$$

$$T = S_{67} + S_{68} + S_{69} = 1.33 \text{ 抽}$$

Time Reversible Markov Chain

- A stationary Markov chain observed backwards in time is also Markovian, i.e.

$$P(X_m = j | X_{m+1} = i, X_{m+2}, X_{m+3}, ...) = P(X_m = j | X_{m+1} = i).$$

In this case, the reversed chain has transition probabilities

$$Q_{ij} \triangleq P(X_m = j | X_{m+1} = i) = \frac{\pi_j P_{ji}}{\pi_i}.$$

If $Q_{ij} = P_{ij}$ for all i, j, we say the Markov chain is time reversible.

- **Proposition:** For a time-reversible chain, its stationary probabilities must satisfy

$$\pi_i P_{ij} = \pi_j P_{ji}.$$

Time Reverse

- If a positive recurrent Markov chain can only make transitions from a state to one of its two nearest neighbors, it is time reversible.


CASIO
fx-991CN X 中文版
CLASSWIZ
1：计算
SHIFT
ALPHA
菜单 设置
开机
SOLVE
OPTN
CALC
STO
ENG
DEL
AC
7
8
9
4
5
6
1
2
3
0
Ans

- 菜单-4：进入矩阵模式
- 定义完矩阵——按AC
- 按OPTN：获取矩阵

- 数学归纳法
- 数列